



AI-driven Optimisation of biodiesel blends for enhanced engine efficiency and reduced emissions

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ABSTRACT

Diesel engines are widely used in industrial applications for their high-power and fuel efficiency. However, concerns over fossil fuel depletion and strict emission regulations have spurred the search for sustainable alternatives. Biodiesel, derived from renewable sources, offers a promising solution due to its lower emissions and compatibility with diesel engines. This study investigates the potential of Lemon Grass Oil (LGO) and Palm Jatropha Oil (PJO) as biodiesel feedstocks, analyzing their performance and environmental impact. LGO and PJO biodiesels were synthesised via transesterification and blended with diesel (80:20). Engine tests were conducted under varying Compression Ratios (CR), Injection Pressures (IP), and Injection Timings (IT). Performance metrics such as Brake Thermal Efficiency (BTE) and Brake Specific Fuel Consumption (BSFC) were evaluated alongside emissions (CO, NO_x, HC, and smoke opacity). PJO-based blends significantly reduced emissions, while both biodiesels exhibited competitive efficiency. A Support Vector Regression (SVR) model optimised engine parameters using the Discrete Grey Wolf Optimization (DGWO) algorithm, achieving a BTE of 30.12% and BSFC of 0.396 kg/kW-hour. These findings highlight the potential of LGO and PJO biodiesel blends as sustainable alternatives, contributing to cleaner transportation and energy-efficient solutions.

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1. Introduction

Diesel engines are growing increasingly viable alternatives in the cutting-edge industrial economy due to their high-power output, greater accessibility, and fuel efficiency. The manufacturing industry can now design lighter, smaller engines with higher power owing to the widespread use of cutting-edge diesel engine technologies, such as split injection approaches, turbocharging techniques, higher pressure direct fuel injection, and emission control STRATEGIES. In addition to worries about the depletion of fossil fuels, the search for alternatives to today's petroleum-based fuels is crucial because they can enable low-carbon or carbon neutral transportation if the fuel is derived from biomass or clean primary energy sources like solar, wind, and hydroelectric power. Biofuels are primarily defined as ethanol, biodiesel, and hydrotreated vegetable oil (HVO). They are typically combined at low levels with fossil fuels, such as the previously stated E10 gasoline and B5 and B20 (biodiesel blends with blending levels of less than 5% and 20%, respectively) (Yue and Liu 2023). Because it can be manufactured using waste and leftover feedstock and has various advantageous attributes as an IC, HVO is thought to be a major driver of the growth in biofuel production. On the other hand, depletion, vibration, and greenhouse gas emissions from these engines continue to be major issues (Gomaa, Mustafa, and Al-Dmour 2020).

Diesel engines emit 30–100 times more particulate matter than gasoline engines, making compliance with increasingly stringent emission standards and regulations crucial (Soudagar et al. 2025a). Furthermore, the worldwide energy crisis has

prompted the development of more exacting evaluation techniques to monitor the practical performance and emission characteristics of diesel engines. Because of this improved awareness of the risks that the environment poses to human health, we are able to regulate engine emissions. Biodiesel's advanced value-adding can yield multiple benefits, including non-toxic fuel substitution, decreased reliance on fossil fuels, reduced pollution emissions, averting or reducing environmental fluctuations, and fostering a circular economy while boosting employment (Soudagar et al. 2025b). Since biodiesel has similar chemical and physical properties to fossil fuels and can be blended in any ratio with petroleum diesel, it is considered one of the clean and sustainable fuels. In addition, it can run engines without the need for modifications (Sateesh et al. 2024) and has a greater flash point (e.g. 128 °C for PJO and 90 °C for LGO) than conventional petroleum diesel (58°C) (Venu et al. 2025). Through chemical processes, biodiesel can be produced from various animal fats (lard, tallow, etc.), raw vegetable oils (corn, palm, etc.), non-edible oils (algae, jatropha, etc.), and vegetable waste oil (Zhang et al. 2021). Because bio-oils are free of sulphur and aromatic compounds, have a higher cetane number, more oxygen, and are effective in lowering diesel engine exhaust emissions. The oxygen concentration of biodiesel fuel varies between 20% and 100% by volume, according to National Renewable Energy Laboratory (NREL) (Mizik and Gyarmati 2022). However, oxygen is not present in petroleum diesel. The biodiesel's significant oxygen molecule content lowers particle emissions and boosts thermal efficiency. Nonetheless, the primary drawback